

ESA's Juice Gains Speed During Earth Flyby Toward Jupiter

From: ScienceDaily All Mined: 2026-10-03 5 min read

The European Space Agency's Jupiter Icy Moons Explorer, or Juice, made a close pass by Earth on September 28. It used our planet's gravity to reshape its path toward Jupiter while consuming very little fuel. This maneuver is called a **gravity assist**.

The carefully planned encounter changed Juice's direction by 20 degrees relative to its previous path and increased its speed by 3.5 kilometers per second. During the four weeks before the flyby, mission controllers needed only one small course correction out of six opportunities set aside. Controllers began particularly close monitoring on August 17, with intensive tracking planned to continue through October 10.

Juice reached its closest point to Earth at 13:45 CEST, or 11:45 UTC, on September 28. The spacecraft passed 8,640 kilometers above the Indian Ocean while taking a sequence of images with its monitoring cameras.

Angela Dietz, Juice's Spacecraft Operations Manager, explained that precise navigation and careful planning kept fuel use low. Using only a small amount of the **propellant** reserved for this flyby leaves more available for observations of Jupiter's icy moons.

Changing the spacecraft's path was the primary purpose of the flyby. The encounter also gave the mission team a chance to test Juice's science instruments using a real planetary target. This was the third such opportunity during the mission. The first came during Juice's 2024 lunar-Earth **gravity assist**, followed by observations of Comet 3I/ATLAS in 2025.

ESA teams spent months coordinating with the outside groups responsible for Juice's 10 scientific instruments. They scheduled instrument activities to make the most of the spacecraft's limited time and resources.

Juice project scientist Claire Vallat, who led the effort, explained that the journey offers few opportunities to calibrate and validate instruments under well-understood conditions. **Calibration** helps check and adjust measurements, while **validation** checks whether instruments work as expected.

Vallat noted that activities sometimes needed priority decisions. For example, Juice relied on battery power alone while in Earth's shadow. During the flyby, the team **prioritized calibration** activities relevant to work at Jupiter. They also considered whether an opportunity was unique or could be scheduled later.

Juice is scheduled to carry out 35 flybys of Jupiter's large moons Ganymede, Callisto and Europa. Discussions about instrument operations for the Europa encounters began about 10 years ago and will continue until the spacecraft passes that moon in the early 2030s.

Earth served as a rehearsal for these more demanding encounters. Scientists could check how the instruments behave in space and refine tools for analyzing future data. Juice also spent several days traveling through Earth's **magnetotail**, the long extension of its magnetic field that stretches away from the Sun.

Word Treasure

- gravity assist** -- A maneuver that uses a planet's gravity to change a spacecraft's path and speed.
- propellant** -- Material a spacecraft uses to produce thrust and adjust its motion.
- calibration** -- Checking and adjusting an instrument's measurements.
- validation** -- Checking whether an instrument works as expected.
- prioritized** -- Gave some activities greater importance when deciding what to do.
- magnetotail** -- The extension of Earth's magnetic field that stretches away from the Sun.

Background you need

Calibration checks and adjusts measurements; validation checks whether instruments work as expected.

A planetary flyby can provide an opportunity to test instruments before later encounters.

Break it down

Introduces the Earth flyby and its gravity assist.

Gives navigation, tracking and closest-approach details.

Explains fuel savings and earlier instrument-testing opportunities.

Describes instrument coordination, calibration and priority decisions.

Connects the rehearsal to planned moon encounters and mentions Earth's magnetotail.

Why it matters

Saving propellant and checking instruments during the Earth encounter helps prepare Juice to gather scientific information during its planned flybys of Jupiter's icy moons.

Test what you remember

1. What was the primary purpose of Juice's Earth flyby?
 - (A) To land beside the Indian Ocean
 - (B) To deliver instruments to Earth
 - (C) To change the spacecraft's path
 - (D) To collect samples from a comet
2. By how much did the gravity assist increase Juice's speed?
 - (A) 4.5 kilometers per second
 - (B) 1.5 kilometers per second
 - (C) 2.5 kilometers per second
 - (D) 3.5 kilometers per second
3. Which event provided an earlier instrument-testing opportunity in 2025?
 - (A) Observations of the moon Ganymede
 - (B) Observations of Comet 3I/ATLAS
 - (C) Observations of the moon Callisto
 - (D) Observations of the moon Europa
4. What does calibration help scientists do?
 - (A) Collect and store more fuel
 - (B) Check and adjust measurements
 - (C) Choose and name new moons
 - (D) Plan and build new cameras
5. Which consideration helped the team prioritize calibration activities?
 - (A) Their relevance to work at Jupiter
 - (B) Their relevance to landing on Earth
 - (C) Their relevance to collecting rocks
 - (D) Their relevance to exploring Saturn
6. When is Juice expected to pass Europa, according to the article?
 - (A) In the late 2030s
 - (B) In the late 2040s
 - (C) In the early 2030s
 - (D) In the early 2040s

Answer key (try the quiz first!): 1: C 2: D 3: B 4: B 5: A 6: C

Questions to think about

1. **Angela Dietz:** She attributed low fuel use to precise navigation and careful planning, explaining that it preserves propellant for observations at Jupiter.
2. **Claire Vallat:** She explained that instrument-testing opportunities are limited and that priorities depend on preparation for Jupiter and whether an opportunity can occur later.

MY THOUGHTS (at least 20 words):